How can AI classify sleep apnea from multimodal data?

Owner: Laurin Engelen

Common 5-step pipeline and the three main model approaches in recent papers

1. Record — Clinic PSG or PM › **2. Clean Data** — Noise & scaling › **3. Cut Windows** — 1s to 60s blocks › **4. Run Model** — Classic ML vs DL › **5. Evaluate** — Scores & heatmaps

Classic ML Pipeline

Hand-Crafted Numbers

- **Calculate stats:** Mean, standard deviation, peak oxygen drop
- **Frequency:** Slow heart rate swings (HRV) and brain wave bands
- **Pick best features:** Select top features via statistical tests (ANOVA)
- **Fast classifiers:** Random Forest, LightGBM, or Support Vector Machines

1D Deep Learning

Direct Raw Time Series

- **Direct input:** Raw normalized sensor waves go straight into neural nets
- **1D-CNN layers:** Spot instant patterns across all sensors together
- **BiLSTM layers:** Learn how signals change step-by-step over time
- **Example (P4 Alarcón):** Compact 6-layer 1D-CNN on 60s windows

2D Image Deep Learning

Signals Turned into Pictures

- **Convert waves:** Turn 1D signals into 2D time-frequency images
- **Methods:** Wavelet Scalograms (CWT) and STFT spectrograms
- **Snoring sound:** Turn audio recordings into Mel-spectrogram pictures
- **Standard vision nets:** Classify using ResNet, VGG, or ShuffleNet (P3)

Sources: P2 Fathima & Ahmed 2025 (p.6–14), P3 Peng et al. 2025 (p.3–6), P4 Alarcón et al. 2023

Multimodal Fusion Strategies in the Literature

Owner: Laurin Engelen

How recent research papers combine multiple physiological signals (P3, P4, P5)

1. Early Fusion

Stack Signals at the Start

[SpO2, Pulse, Belts] ➔ One single matrix (Channels × Time) ➔ One model

- Resample all sensors to a shared speed (1 Hz in P4)
- Model sees signal links right away (pulse jump + O2 drop)
- Used in P4 Alarcón 2023 (4 channels ➔ **84.3% Accuracy**)

2. Feature-Level Fusion

Separate Sub-Branches First

[Airflow] ➔ Branch 1
[SpO2] ➔ Branch 2 ➔ Merge with Attention

- Each sensor keeps its own natural frequency (Hz)
- Attention layers decide which sensor to trust most
- Example (P3 Peng 2024): Airflow + SpO2 reaches **95.9% Acc**

3. Late Decision Fusion

Combine Final Predictions

Model 1 (SpO2) + Model 2 (ECG) ➔ Weighted average vote

- Train a completely separate model for every sensor
- Average the prediction scores at the very end
- Keeps working even if one body sensor slips off in bed

Key Finding & Research Gap (P5 Salvi et al. 2024): Combining sensors always outperforms single sensors (+5% to +15% accuracy boost). However, current sleep studies almost exclusively fuse biosignals and rarely include clinical patient data like Age, BMI, or Sex.

Sources: P3 Peng et al. 2025 (p.7–16), P4 Alarcón et al. 2023 (p.4–9), P5 Salvi et al. 2024 (p.1–10)

Signal-Specific Data Pipelines in SOTA Papers

Owner: Laurin Engelen

How different body signals are cleaned, split into windows, and modeled (P1–P4)

Signal & Source Paper	Cleaning & Filtering	Window Length & Split	What Goes Into Model?	Top Model & Score
SpO2 + Pulse + Belts P4 Alarcón et al. 2023	• Downsample all to 1 Hz • Fix bad values (O2 < 80%) • Z-Score normalization	• Clean 60s windows • Event ≥ 10s ➔ Apnea (1) • Balanced 50:50 training	• Raw 1 Hz signal matrix • Preserves exact seconds for Grad-CAM heatmaps	• **1D-CNN + GAP** • 84.3% Accuracy (92.1% AUC on SHHS2)
Airflow & SpO2 P3 Peng et al. 2025	• 0.05–3 Hz gentle filter • Clip unreasonable drops • Min-Max or Z-Score	• 10s, 30s, or 60s windows • Loss functions tuned for rare events	• Direct 1D breathing waves • Oxygen desaturation depth (≥ 3% drop index)	• **Dual-Branch CNN** • 95.9% Accuracy with Channel Attention
ECG Pulse & Snoring P3 Peng et al. 2025	• 0.5–40 Hz filter for ECG • Remove baseline sway • Audio noise filtering	• 60s windows for ECG (fits 30–90s apnea loop) • 20–40 ms audio frames	• R-peak intervals (HRV VLF) • Derived breathing (EDR) • Sound pictures (Mel-spec)	• **CNN-LSTM** (97.2%) • SVM / Random Forest (up to 99.1%)
Brain Waves (EEG) P2 Fathima & Ahmed 2025	• Remove 50 Hz power hum • 0.5–40 Hz standard filter • Right side C4-A1 works best	• Short 1s to 10s slices (keeps fast brain signals stable & reliable)	• Energy in 6 brain bands • Beta/Delta power ratio • Signal complexity values	• **BiLSTM / GRU** • 92.7% to 99.3% Acc for arousal detection

Sources: P1 Gaiduk et al. 2023, P2 Fathima & Ahmed 2025, P3 Peng et al. 2025, P4 Alarcón et al.

Key Pipeline Challenges & Insights in SOTA

Owner: Laurin Engelen

Crucial methodological decisions and pitfalls identified across papers P1 to P6

1. Scaling & Hospital Shift

- **Scaling is critical (P4):** Without Z-Score normalization, 1D-CNN accuracy collapses from **84.3% down to 72.3%** on SHHS2.
- **Hospital differences (P3):** Models trained on one hospital drop by over 12% on external clinics due to differing sensor hardware.

2. Window Length & Fast Signals

- **Why 60s windows (P4):** Perfectly captures full apnea events plus the delayed 15–45s blood oxygen drop.
- **Fast signals need sub-slices (P2):** Fast brain waves (EEG) become unstable over 60s and require 1–10s sub-frames.

3. Data Imbalance & Leakage

- **Unbalanced data (P3, P4):** Normal breathing dominates (~80%). Solved via 50:50 undersampling or Focal Loss.
- **Strict patient separation (P1):** Never mix windows of the same person across train and test (artificially inflates scores by 10%).

4. False Alarms & Explanations

- **The wake-up problem (P4):** Normal movements while awake cause false alarms if divided by 8h recording time instead of true sleep.
- **Explainable AI (P4 & P6):** 1D-Grad-CAM heatmaps highlight exact seconds so doctors understand why an apnea was detected.

Sources: P1 Gaiduk 2023 (p.15–20), P2 Fathima 2025 (p.16–20), P3 Peng 2025 (p.16–19), P4 (p.15–18)

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Core Strategy & Model Choice

1. HARMONIZATION **1 Hz Feature Alignment**

- Bring all signals to 1 value per second (1 Hz).
- Fast signals (EEG/ECG) are converted into 1-second energy values.
- Every 60-second window gets the exact same size.

2. FUSION **Modular Sensor-Toggle**

- Turn individual sensors ON or OFF in code.
- Easy to test: Which sensors are really needed?
- Matches different devices (from simple smartwatches to sleep labs).

3. MODEL **1D-CNN Baseline (P4 Alarcón)**

- A simple neural network that reads time series directly.
- Detects if an apnea event is happening (Yes / No).
- Shows heatmaps (Grad-CAM) to explain *why* the model made a decision.

Target Output	`0 = No Event, 1 = Apnea`

Pipeline Architecture

1. **Data Acquisition & Adapters**
   Load files (SHHS, UCDDB, ISRUC) and give all sensorik signals standard names.
2. **Pre-processing & Artifact Filtering**
   Fix wrong values (e.g. SpO2 < 70% or broken heart rate readings).
3. **Windowing & Labeling**
   Overlapping 60s windows; label = 1 if annotated event ≥ 10 s.
4. **Model / Classification**
   The 1D-CNN scans the 60-second window and predicts the probability of an apnea.
5. **Evaluation & Explainability**
   Window metrics (Acc, Sens, Spec, LR+) + 1D-Grad-CAM visualization.

Sources: P3 (p.3–4), P4, P5

Dynamic Sensor-Toggle & Clinical Presets

Owner: Laurin Engelen

WEARABLE — 2 Signals

1. Finger Only

`SpO2 + Pulse (HR)`

- ✓ Simple & comfortable (smartwatch level)
- ✗ Oxygen drops 15–45s late; misses short events

Paul et al. 2022 Expected

~78 – 82%

MEDIUM — 4 Signals

2. Home Monitor

`+ Chest & Belly Belts`

- ✓ **Setup from Alarcón et al. (Paper P4)**
- ✓ Belts show breathing stops with zero delay

Paper P4 Result (SHHS2)

84.3% Accuracy

EXTENDED — 5 Signals

3. Add Airflow

`+ Nose Airflow (FLOW)`

- ↗ **Biggest performance jump (+5–10%)**
- ✓ Directly shows if air stops or keeps flowing

Paper P3 Result (Peng 2025)

~90 – 94%

SLEEP LAB — 6–7 Signals

4. Full Lab Setup

`+ Brain (EEG) & ECG`

- ✓ Detects wake-up reactions in the brain
- ✓ Prevents false alarms while patient is awake

Paper P2 Result (Fathima 2025)

~93 – 96%

Sources: P4 Alarcón et al. 2023 (p.4-5), Collop et al. 2011 (AASM Guidelines)